Two complementary sources of evidence for variant effects

GRCh38 SNVs | HIGH-impact excluded | Odd chromosomes → train/validation · Even chromosomes → test

Mendelian disease

CLINICAL EVIDENCE — TraitGym-inspired

- **Pathogenic** — OMIM + Smedley et al. + HGMD · AF < 0.1%
- **Control** — gnomAD non-rare SNVs · AF > 0.1%

1 pathogenic + 9 controls
Matched on chromosome · consequence
TSS distance · exon distance

SUBSET COUNTS — Dark: pathogenic · Tint: controls · Total

Missense	10,000
Splicing	5,500
5′ UTR	3,050
Promoter	2,700
ncRNA	1,590
3′ UTR	1,100
Distal	990
Synonymous	650
Mature miRNA	50

25,630 variants
2,563 pathogenic · 23,067 controls

Saturation genome editing

EXPERIMENTAL EVIDENCE — MaveDB · 11 studies

CRISPR-HDR variant library
Cell fitness / proliferation → continuous score
Only Missense + near-exon Splicing retained

Calibrated class → abnormal vs normal
ClinGen / ExCALIBR or author thresholds

SUBSET COUNTS — Dark = abnormal · Tint = normal

Missense	31,316 total	4,250 abnormal · 27,066 normal
Splicing	7,425 total	735 abnormal · 6,690 normal

No matching · no subsampling
Macro-averaged AUPRC within each MaveDB study

38,741 variants
4,985 abnormal · 33,756 normal

Relevant training region — CDS · Upstream · Downstream · ncRNA · Enhancer